

detached cache

statistics cache
 q^{in}, q^{out}, q^R
 $\hat{d}, \hat{o}, \gamma, \tau$

read only

RLB matrix lane: MatrixPolicy-updated A_l, B_l

MatrixPolicy

group
gates

matrix gradi
scaling

amW matrix
branch

transient
Muon

pair rescale
every 5 steps

updated
 A_l, B_l

$$A_g \leftarrow e^\ell A_g, B_g \leftarrow e^{-\ell} B_g$$

statistics only

AdamW-only bypass lane

AdamW-only parameters
rational coefficients, attention, embeddings, norms

AdamW
only

updated
 $\theta_{\neg A, B}$

excluded from MatrixPolicy
no gates, Muon, rescale

— MatrixPolicy-updated RLB matrices — AdamW-only parameters